

markpact

Live Contract Generation

PROMPT: calculator-api

*Kalkulator REST API. FastAPI. Endpoints: POST /calculate z {operation, a, b}. Obsługa +, -, *, /, sqrt, pow*

Model: ollama/qwen2.5-coder:14b

1. Generowanie kontraktu

LLM: ollama/qwen2.5-coder:14b

README.MD

```
# Kalkulator REST API

A simple calculator REST API built with FastAPI that supports basic arithmetic operat

## Features
- Perform addition (+)
- Perform subtraction (-)
- Perform multiplication (*)
- Perform division (/)
- Calculate the square root (sqrt)
- Calculate power (pow)

## API Endpoints
- POST /calculate - Performs a calculation based on provided operation, a, and b

---

```text markpact:deps python
fastapi
uvicorn
pydantic
```

```python markpact:file path=app/main.py
from fastapi import FastAPI, HTTPException
from pydantic import BaseModel
import math

class CalculationRequest(BaseModel):
 operation: str
 a: float
 b: float = None # b is optional for sqrt and pow operations

app = FastAPI()

@app.get("/health")
def health_check():
 return {"status": "ok"}

@app.post("/calculate")
def calculate(request: CalculationRequest):
 operation = request.operation.lower()
 a = request.a
 b = request.b if request.b is not None else a # Use a for sqrt and pow if b is n

 try:
 if operation == '+':
 return {"result": a + b}
 elif operation == '-':
 return {"result": a - b}
```

WALIDACJA + OCZEKIWANIA

Prompt: calculator-api

Kalkulator REST API. FastAPI. Endpoints: POST /calculate z {...

Model: ollama/qwen2.5-coder:14b

Czas: 110130ms

Rozmiar: 2424 znakow

Linie: 82

WALIDACJA

[V]

LLM odpowiedzial

110130ms

[V]

README.md zapisany

generated/live/README.md

## 2. Parsowanie blokow

parse\_blocks() -> 4 blokow

README.MD

```
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WALIDACJA + OCZEKIWANIA

Znaleziono 4 blokow:

```
markpact:deps (24 chars)
markpact:file=app/main.py (1269 chars)
markpact:run (65 chars)
markpact:test (480 chars)
```

WALIDACJA

[V]

markpact:deps

lang=text, 24 chars

[V]

markpact:file=app/main.py

lang=python, 1269 chars

[V]

markpact:run

lang=bash, 65 chars

[V]

markpact:test

lang=text, 480 chars

3. Walidacja blokow

Sprawdzenie wymaganych blokow markpact:*

README.MD

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WALIDACJA + OCZEKIWANIA

Wymagane bloki:

markpact:deps

-- zaleznosci

markpact:file

-- kod zrodlowy

markpact:run

-- komenda uruchomienia

Opcjonalne:

markpact:test

-- testy HTTP

markpact:target

-- platforma docelowa

Wynik: PASS

WALIDACJA

[V]

markpact:deps obecny

wymagany

[V]

markpact:file obecny

wymagany

[V]

markpact:run obecny

wymagany

[V]

markpact:test obecny

opcjonalny

[?]

markpact:target

opcjonalny, brak

4. Analiza zawartosci

Szczegolowa analiza kazdego bloku

README.MD

```
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WALIDACJA + OCZEKIWANIA

Analiza blokow:

Deps: 3 pakietow

fastapi, uvicorn, pydantic

File: app/main.py (40 linii)

Run: uvicorn app.main:app --host 0.0.0.0 --port \${MARKP

Test: 7 assercji HTTP

WALIDACJA

[V] 3 zaleznosci

fastapi, uvicorn, pydantic

[V] file=app/main.py

40 lines, sha256=9f685f803975

[V] Run command

uvicorn app.main:app --host 0.0.0.0 --port \${MARKPACT_PORT:-

[V] 7 testow HTTP

GET/POST/PUT/DELETE

5. Integralnosc SHA-256

Hash kazdego bloku markpact:*

README.MD

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```

WALIDACJA + OCZEKIWANIA

Integralnosc danych:
README.md: bed014a73c6e62404559363635a89396

markpact:deps: 1ef577a673407cd0
markpact:file=app/main.py: 9f685f803975437c
markpact:run: b49cce15b5712fdd
markpact:test: 09a96da7f4c2379a

Kazdy blok ma unikatowy hash.
Zmiana = nowy hash = wykryta.

WALIDACJA

[V] README.md hash
bed014a73c6e624045593636

[V] markpact:deps
sha256=1ef577a673407cd0

[V] markpact:file=app/main.py
sha256=9f685f803975437c

[V] markpact:run
sha256=b49cce15b5712fdd

[V] markpact:test
sha256=09a96da7f4c2379a

Podsumowanie

4

blokow

markpact:*

10

passed

walidacja

0

failed

bledy

110.2s

czas

total

Kroki:

- [v] LLM generate_contract() (110130ms)
ollama/qwen2.5-coder:14b, 110130ms, 2424 chars
- [v] parse_blocks() (0ms)
4 blocks
- [v] validate markpact:deps
present
- [v] validate markpact:file
present
- [v] validate markpact:run
present
- [v] Analiza deps
3 packages: fastapi, uvicorn, pydantic
- [v] Analiza file=app/main.py
40 lines, sha=9f685f803975
- [v] Analiza run
uvicorn app.main:app --host 0.0.0.0 --port \$MARKPACT_PORT:8000
- [v] Analiza test
7 assertions
- [v] SHA-256 README.md
bed014a73c6e624045593636

Kontrakt wygenerowany z 1 promptu -> 4 blokow markpact